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Consulting Proposal
House of Blanks - Bottleneck Analysis

Prepared For
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1. Project Executive Summary

This project was developed for House of Blanks, a factory brand under Roopa Knitting Mills, a leading producer of premium knit fabrics and apparel. The analysis focuses on two high-volume, high-profit products: 1008MWT (1008 Midweight T-Shirt) and 1011HWLS (1011 Heavyweight Long-Sleeve T-Shirt), which together generate more than 70% of the profit within their category for the company. Throughout this report, both items will be referred to by their SKU identifiers.

To understand current and future production requirements, three years of monthly demand were analyzed to identify trends, seasonal patterns, and projected workload. This information was used to establish the output levels that the sewing line must consistently meet to avoid shortages and protect profitability.

A detailed review of the sewing line was then completed to evaluate operator flow, cycle times, and machine utilization. During this assessment, a significant buildup of work-in-process was observed at the Close Binding + Label operation. Time-study data confirmed that this station operates at a slower rate than the rest of the line, making it the primary bottleneck in the process.

This bottleneck limits the line’s ability to meet demand, causes idle time at downstream stations, and introduces unnecessary congestion between operations. Resolving the delay at the Close Binding + Label step is essential to improving flow, increasing throughput, and aligning production capacity with the demand levels identified in this study.

2. Forecast Data & Analysis

2.1 Annual demand

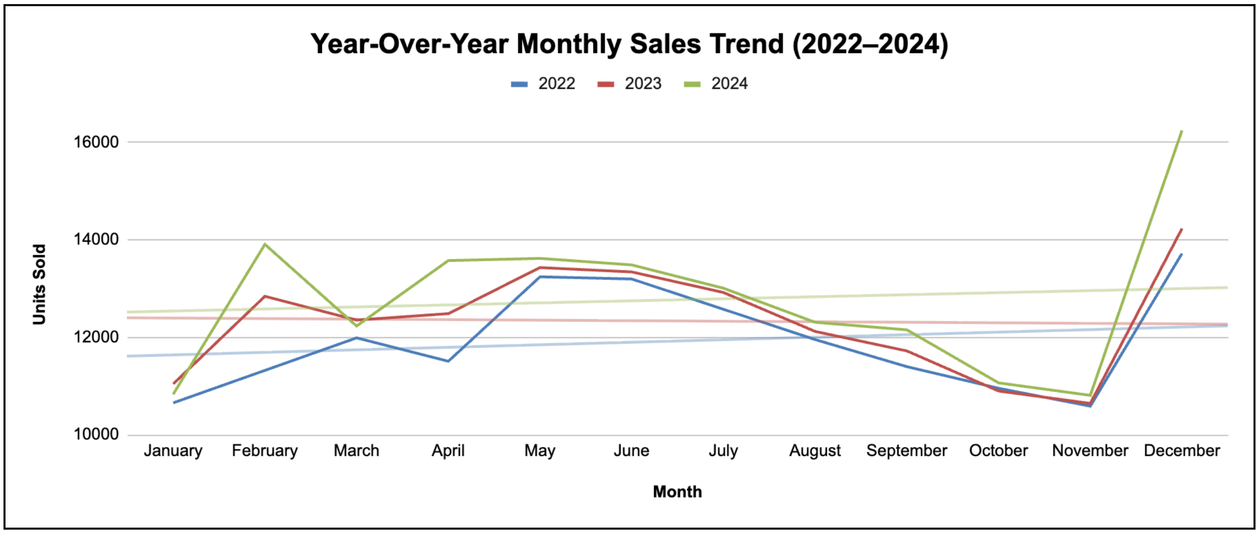
1008MWT

Prior demand data for 1008MWT

Month	2022	2023	2024	Total Demand
January	10,655.000	11,044.000	10,833.000	32,532.000
February	11,322.000	12,844.000	13,910.000	38,076.000
March	11,990.000	12,355.000	12,233.000	36,578.000
April	11,510.000	12,488.000	13,577.000	37,575.000
May	13,244.000	13,433.000	13,622.000	40,299.000
June	13,199.000	13,344.000	13,488.000	40,031.000
July	12,577.000	12,922.000	13,010.000	38,509.000
August	11,955.000	12,122.000	12,310.000	36,387.000

September	11,399.000	11,722.000	12,155.000	35,276.000
October	10,955.000	10,899.000	11,066.000	32,920.000
November	10,588.000	10,644.000	10,810.000	32,042.000
December	13,722.000	14,234.000	16,249.000	44,205.000
				444,430.000

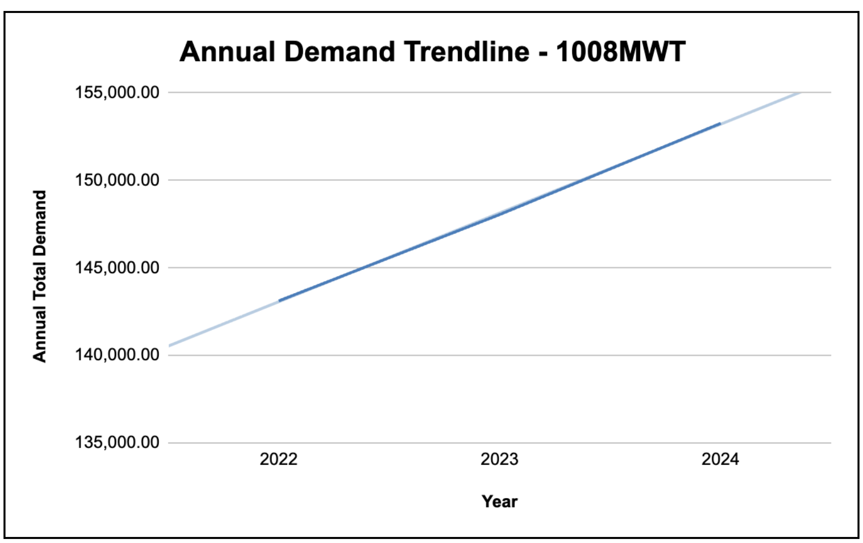
Year-over-year monthly sales trend for 1008MWT



Annual total demand for 1008MWT

Year	Annual Total Demand
2022	143,116.00
2023	148,051.00
2024	153,263.00

Annual demand trendline for 1008MWT



The 1008 Midweight T-shirt shows strong and consistent growth across the three-year period analyzed. Total annual demand increased from 143,116 units in 2022, to 148,051 in 2023, and 153,263 in 2024, confirming a clear upward trajectory for one of House of Blanks' most important products.

Monthly demand patterns show predictable spikes in April through July and again in December. These peaks make sense for House of Blanks because the company serves both B2B wholesale clients (print shops, streetwear brands, merch companies) and B2C retail customers.

- Spring/summer peaks likely come from brands preparing for summer collections, event merch, festivals, and increased consumer apparel spending as temperatures rise.
- December demand is characteristic of wholesale restocks heading into Q1, plus B2C holiday shopping.

Operating as both B2B and B2C gives House of Blanks more stable demand overall, but it also creates layered seasonality that needs to be reflected in forecasting and production planning.

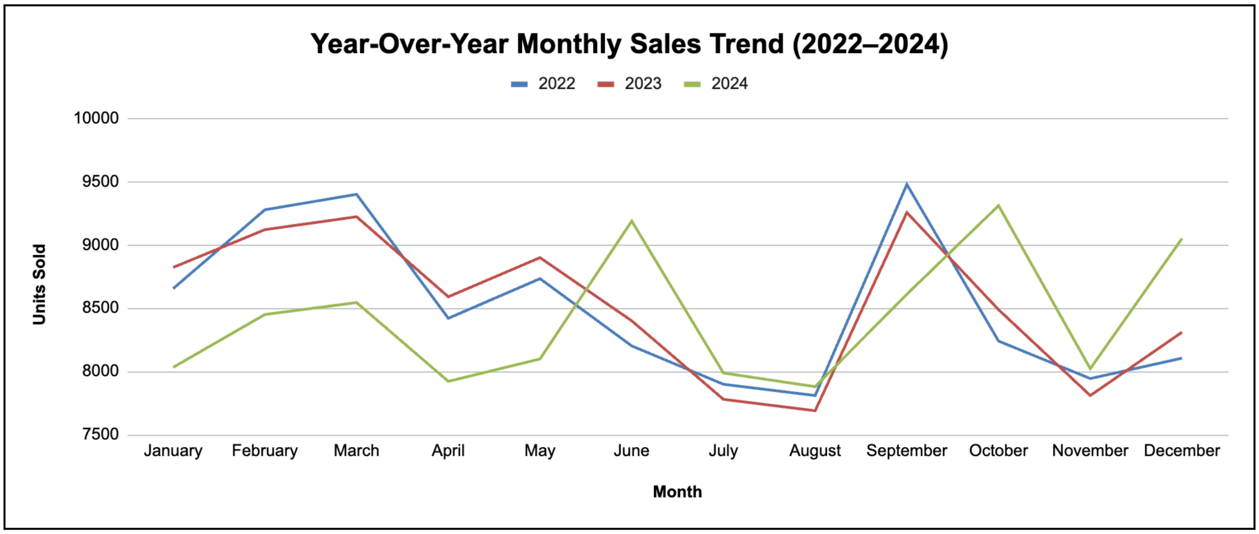
Overall, the 1008MWT continues to grow steadily, reinforcing the need for aligned production capacity and detailed inventory planning to support rising volume.

1011HWLS

Prior demand data for 1011HWLS

Month	2022	2023	2024	Total Demand
January	8,655.000	8,822.000	8,033.000	25,510.000
February	9,277.000	9,120.000	8,450.000	26,847.000
March	9,399.000	9,222.000	8,544.000	27,165.000
April	8,420.000	8,590.000	7,922.000	24,932.000
May	8,733.000	8,899.000	8,099.000	25,731.000
June	8,202.000	8,400.000	9,188.000	25,790.000
July	7,899.000	7,780.000	7,988.000	23,667.000
August	7,810.000	7,690.000	7,880.000	23,380.000
September	9,477.000	9,255.000	8,610.000	27,342.000
October	8,240.000	8,488.000	9,310.000	26,038.000
November	7,944.000	7,810.000	8,022.000	23,776.000
December	8,105.000	8,310.000	9,050.000	25,465.000
				305,643.000

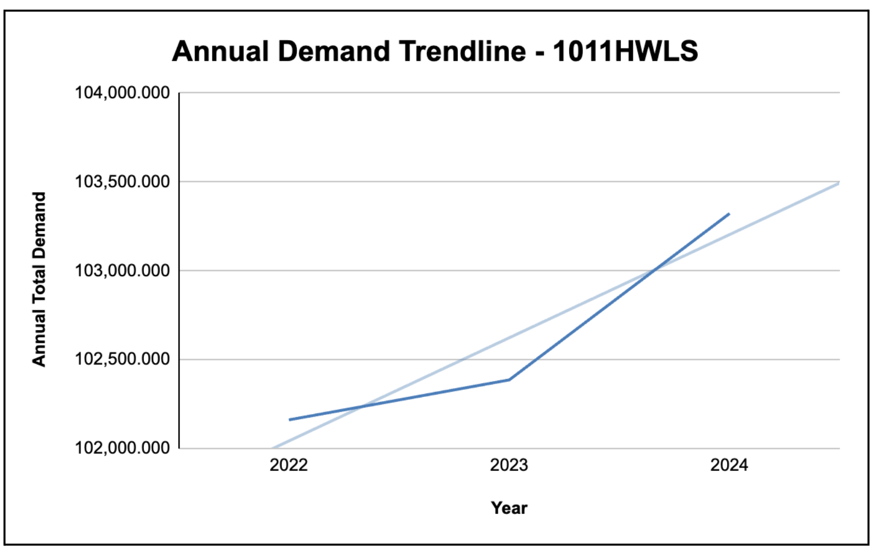
Year-over-year monthly sales trend for 1011HWLS



Annual total demand for 1011HWLS

Year	Annual Total Demand
2022	102,161.000
2023	102,386.000
2024	103,323.000

Annual demand trendline for 1011HWLS



Annual demand for the 1011 Heavyweight Long-sleeve T-shirt has remained steady across the three years reviewed, increasing slightly from 102,161 units in 2022 to 103,323 units in 2024. This indicates a mature product with stable, predictable volume.

The monthly pattern is consistent year over year: modest strength in the first quarter, a dip through the summer, and a recovery in the fall. These shifts align naturally with weather-driven buying behavior and the long-sleeve nature of the product. Unlike the 1008 Midweight T-shirt, this SKU shows less variability and fewer pronounced seasonal spikes.

2.2 Forecast analysis

Seasonal Index Calculation for 1008MWT

Month	Avg of 2022–2024	Seasonal Index
January	10,844.000	0.878
February	12,692.000	1.028
March	12,192.667	0.988
April	12,525.000	1.015
May	13,433.000	1.088
June	13,343.667	1.081
July	12,836.333	1.040
August	12,129.000	0.982
September	11,758.667	0.952
October	10,973.333	0.889
November	10,680.667	0.865
December	14,735.000	1.194

Seasonal Index Calculation for 1011HWLS

Month	Avg of 2022–2024	Seasonal Index
January	8,503.333	0.994
February	8,949.000	1.046
March	9,055.000	1.059
April	8,861.333	1.036
May	8,577.000	1.003
June	8,788.333	1.028
July	7,889.000	0.922
August	7,793.333	0.911
September	9,114.000	1.066
October	8,679.333	1.015
November	7,925.333	0.927
December	8,488.333	0.993

	1008MWT	1011WLS
Lo	11,926.333	8,513.000
To	-36.194	-6.507
α	0.3	0.3
β	0.1	0.1
γ	0.3	0.3

To build a reliable forecast for both products, we used a Holt-Winters multiplicative model, which accounts for level, trend, and seasonality. The model begins by establishing two baseline values:

- Lo (Initial Level):
  This represents the starting estimate of the product's underlying demand before trend or seasonality are applied. For example, the Midweight T-shirt begins with a Lo of about 11,926 units, while the Heavyweight L/S begins with 8,513 units, reflecting their very different demand scales.
- To (Initial Trend):
  This captures the early direction and slope of demand growth or decline.
  A negative To indicates that demand in the baseline period was trending downward; a positive To indicates early upward movement.

The smoothing constants used in the model, α (alpha), β (beta), and γ (gamma), control how quickly the forecast reacts to new data:

- α (level smoothing, set to 0.3): determines how heavily the model weights the most recent actual demand when updating the level.
- β (trend smoothing, set to 0.1): controls how slowly or quickly the trend adjusts; a lower β makes the trend more stable.
- γ (seasonality smoothing, set to 0.3): determines how strongly the model updates seasonal patterns as new months roll in.

Seasonal indices were calculated by averaging monthly demand across 2022–2024 and dividing each month by the overall average. Months above 1.00 represent periods of strong seasonal demand; months below 1.00 represent softer periods. These indices were then fed into the Holt-Winters model so that forecasts reflect real seasonal patterns instead of flat averages.

Once the model structure was set, a full month-by-month forecast was generated for both SKUs. The complete calculations, showing level, trend, seasonal factors, forecast values, and errors for every month, are included in the Data Appendix.

To assess forecast quality, three standard error metrics were calculated:

- MAD (Mean Absolute Deviation):
  Measures the average size of errors without considering direction. Lower values mean tighter accuracy.
- MSE (Mean Squared Error):
  Penalizes larger errors more heavily. It helps evaluate how stable the model is across the entire horizon.
- MAPE (Mean Absolute Percentage Error):
  Expresses errors as a percentage, making it easier to compare performance across products with different demand volumes.

Both products generated low and consistent MAD and MAPE values, indicating the model is performing well and producing forecasts that closely track actual demand. This provides a reliable basis for downstream decisions such as inventory planning, EOQ calculation, staffing, and capacity alignment.

3. Inventory Analysis

This section evaluates the inventory requirements for the two high-volume products, 1008MWT and 1011HWLS, using both deterministic and stochastic approaches. The objective is to determine optimal order quantities, reorder points, safety stock, and total inventory costs under different assumptions about demand variability.

<u>3.1 Parameter Summary</u>

The following table summarizes the key cost and operational parameters used in both inventory models:

Parameter	**1008MWT**	**1011HWLS**	**Notes**
(D) Annual Demand	153,263.000	103,323.000	
(c) Unit Cost	3.900	5.500	
(i) Holding Rate	0.250	0.250	
(h) Holding cost	0.975	1.375	
(K) Setup/ordering cost	200.000	200.000	
(L) Lead Time	10.000	10.000	2 weeks/5 working days
(Wd) Working days	250.000	250.000	Annually

These inputs are based on the annual forecast generated in Section 2, the unit production costs provided by the client, standard industry holding rates, and a two-week replenishment lead time.

3.2 Deterministic Approach

The deterministic model assumes:

- Demand is constant and known with certainty.
- Lead time does not vary.
- No shortages occur.
- Costs remain stable over time.

Under these assumptions, the classical Economic Order Quantity (EOQ) model is applied to determine the optimal order size, expected order frequency, and the associated annual cost components. The reorder point is computed as the demand expected during the fixed replenishment lead time.

Deterministic Approach			
EOQ & policy table	**1008MWT**	**1011HWLS**	**Notes**
EOQ*	7,930	5,482	Units
Orders per year	19.328	18.846	
Time between orders	12.934	13.265	
Average inventory	3,964.755	2,741.241	
Annual setup cost	$3,865.64	$3,769.21	
Annual holding cost	$3,865.64	$3,769.21	
Total relevant cost	$7,731.27	$7,538.41	
Daily demand	613	413	
Reorder point	6,131	4,133	

For both SKUs, the deterministic EOQ model results in stable order quantities and predictable order cycles. The resulting reorder points are driven entirely by average daily demand and a fixed 10-day lead time. While this approach minimizes cost under perfect certainty, it does not account for real-world fluctuations in demand that may lead to stockouts.

3.3 Stochastic Approach

To better reflect real operational conditions, a stochastic model is developed by incorporating historical demand variability. Monthly demand data from 2022 to 2024 are used to estimate the standard deviation of demand, which is then converted into daily variability and lead-time variability. A 95% cycle-service level is selected, corresponding to a z-value of 1.645.

Safety stock is calculated as $SS = z \cdot \sigma L$ and the stochastic reorder point becomes $ROPstoch = dL + SS$.

The EOQ does not change under variability; however, total average inventory and annual holding cost increase due to the addition of safety stock.

Stochastic Approach			
Settings	**1008MWT**	**1011HWLS**	**Notes**
(SL) Service Level	0.95	0.95	95%
z-value	1.645	1.645	95% cycle-service level
σ_month	1272.372	591.225	
σ_day	278.763	129.531	
σ_L	881.525	409.613	
Inventory Policy & Cost			
EOQ*	7,930	5,482	
Avg cycle inventory	3965	2741	
(SS) Safety stock	1450	674	
(ROP) Reorder Point	7,581	4,807	
Total avg inventory	5415	3415	
Annual holding cost	$5,279.49	$4,695.70	
Annual setup cost	$3,865.64	$3,769.21	
Total relevant cost	$9,145.13	$8,464.90	

Compared to the deterministic case, both products experience an increase in reorder point and average inventory. For 1008MWT, the reorder point rises from 6,131 units to 7,581 units, adding 1,450 units of safety stock. A similar effect occurs for 1011HWLS. Although this increases inventory-holding cost, it significantly reduces the probability of stockouts and provides more reliable material availability for production planning.

3.4 Summary of Findings

The deterministic EOQ model provides cost-efficient ordering policies but underestimates needed inventory in a variable-demand environment.

The stochastic model introduces safety stock based on observed historical variability, resulting in higher service levels and reduced risk of disruptions.

For both products, the stochastic reorder point is substantially higher, which strengthens production continuity, particularly important given the bottleneck constraints identified in the sewing line.

4. Operations Analysis

This section evaluates the current production system for the sewing line used to produce both 1008MWT (Midweight T-Shirt) and 1011HWLS (Heavyweight Long-Sleeve T-Shirt). Both products follow the same sequence of operations and use the same equipment, allowing the analysis to focus on a single unified process. The goal is to identify production requirements, assess workforce needs, evaluate the system's control mechanism, and verify the presence of bottlenecks using queueing-theory concepts.

4.1 Process Diagram

The sewing line consists of ten sequential operations beginning with shoulder joining and ending with sleeve tacking. Observations show that material flows in a linear progression with no parallel branches, and all operators work in a paced environment where work arrives from the upstream station.

The full process flow is shown in Diagram 1, and the standard operation times used in the analysis are summarized in Table A.

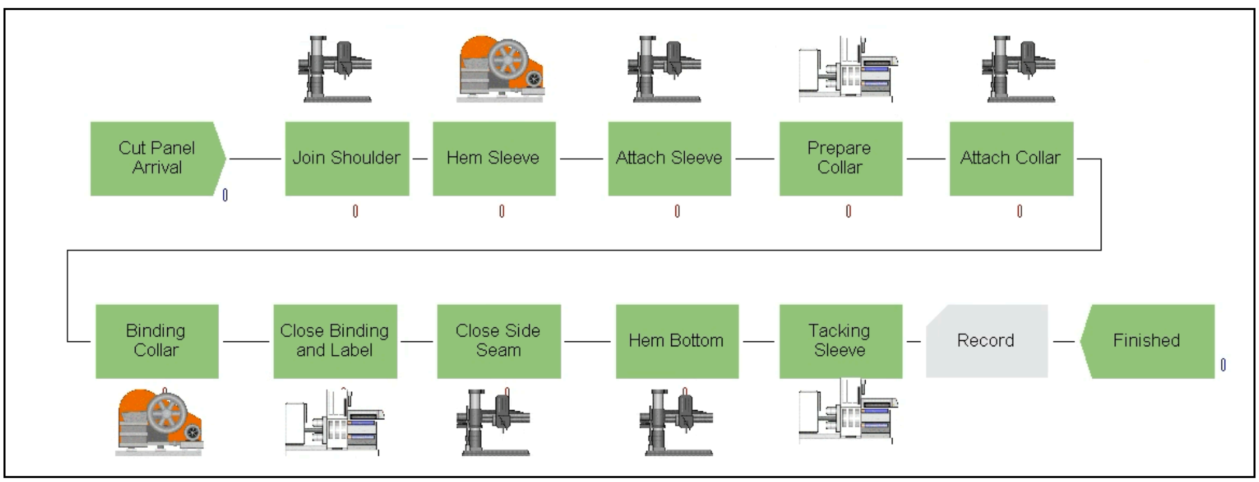


Diagram 1

SN	Operation Name	Machine Type	Standard Time (min)
1	Join Shoulder	O/L	1.2
2	Hem Sleeve	F/L	0.8
3	Attach Sleeve	O/L	0.8
4	Prepare Collar	SNLS	0.45
5	Attach Collar	O/L	0.6
6	Binding Collar	F/L	0.8

7	Close Binding + Label	SNLS	1.3
8	Close Side Seam	O/L	1.4
9	Hem Bottom	O/L	0.9
10	Tacking Sleeve	SNLS	1

Table A

This process structure, serial, single-path, no parallel redundancy, makes the line highly sensitive to its slowest station, which is essential for identifying the bottleneck.

4.2 Production Requirements

To determine the required production capacity, the unified operation times are converted into station capacities and then compared with the forecasted demand for the higher-volume 1008MWT SKU. This ensures the line is capable of meeting the upper-bound load it must support.

The calculated workstation capacities, units-per-hour output, and implied daily requirements are shown in Table B.

Op. #	Operation	Symbol	Time per unit (min)	Capacity per machine (units/hr)	Machines (current)	Machines (proposed)
1	Join Shoulder	JoinShoulder_Op	1.200	50.0	1	1
2	Hem Sleeve	HemSleeve_Op	0.800	75.0	1	1
3	Attach Sleeve	AttachSleeve_Op	0.800	75.0	1	1
4	Prepare Collar	PrepareCollar_Op	0.450	133.3	1	1
5	Attach Collar	AttachCollar_Op	0.600	100.0	1	1
6	Binding Collar	BindingCollar_Op	0.800	75.0	1	1
7	Cls Bind + Lbl	CloseBindingLabel_Op	1.300	46.2	1	2
8	Cls Side Seam	CloseSideSeam_Op	1.400	42.9	1	2
9	Hem Bottom	HemBottom_Op	0.900	66.7	1	1
10	Tacking Sleeve	TackingSleeve_Op	0.900	66.7	1	1

Table B

Op. #	Time per unit (min)	Capacity per station Current (units/hr)	Capacity per station Proposed (units/hr)	Utilization Current	Utilization Proposed
1	1.2	50	50	0.86	1
2	0.8	75	75	0.57	0.67
3	0.8	75	75	0.57	0.67
4	0.45	133.33	133.33	0.32	0.38
5	0.6	100	100	0.43	0.5
6	0.8	75	75	0.57	0.67
7	1.3	46.15	92.31	0.93	0.54
8	1.4	42.86	85.71	1	0.58
9	0.9	66.67	66.67	0.64	0.75
10	0.9	66.67	66.67	0.64	0.75

Table C

Based on these requirements and the capacity analysis presented in Table C, the following conclusions were drawn:

- A minimum of 10 operators are required under the current configuration (one per workstation).
- To meet annual demand, the line must consistently produce:
  - 1008MWT: 153,263 units/year and 613 units/day
  - 1011HWLS: 103,323 units/year and 413 units/day
- (Assuming 250 working days per year.)
- The current system cannot meet the required output, because the Close Binding + Label (Op. 7) workstation has a capacity of only 46.2 units/hr, making it the primary bottleneck.
- A secondary constraint is visible at Close Side Seam (Op. 8), with a capacity of 42.9 units/hr, which also contributes to delays at higher daily demand levels.
- Therefore, additional machines or rebalancing is necessary before the line can meet demand.

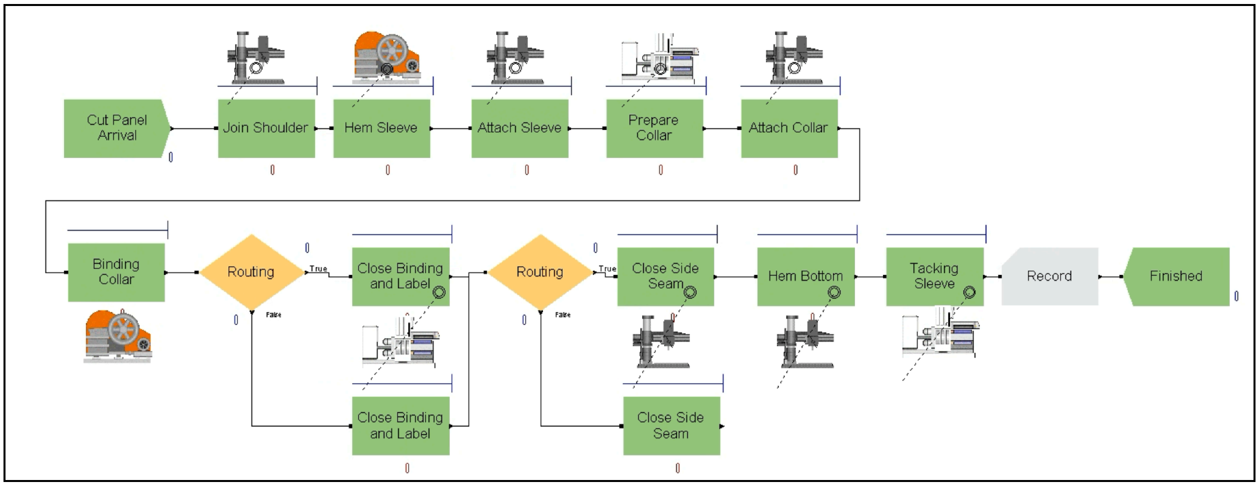


Diagram 2

Diagram 2 illustrates the balanced line configuration after addressing the two major bottlenecks identified in the unbalanced system. By adding one additional machine at Close Binding + Label and one additional machine at Close Side Seam, the capacity of the slowest operations is increased to match the throughput of the rest of the line. This adjustment aligns with the approach recommended in the literature, where parallel resources are added when a single workstation constrains overall output.

In this balanced layout, work is routed to the parallel stations using a simple branching decision rule so that each unit flows to the next available machine. Here, parallelization reduces waiting time, increases flow uniformity, and raises effective line capacity.

With these changes, the production system is capable of meeting the required daily throughput for both SKUs under the forecasted demand. Additionally, balancing the line reduces WIP accumulation before the previously overloaded stations and creates a more stable and predictable workflow, thereby improving both labor utilization and operational performance.

4.3 Evaluation of the Current Production System

4.3.1 Push vs. Pull Determination

The sewing line functions as a push system. Work is released into the line based on upstream cutting schedules rather than real-time operator readiness. There is no kanban mechanism limiting WIP, which results in accumulation of partially sewn garments, especially before the bottleneck station.

Indicators of a push system observed (and supported by queueing behavior):

- Large WIP piles upstream of the bottleneck.
- No card-based or signal-based control.
- Supervisors “dispatch” work rather than operators “requesting” work.

A pull system would reduce WIP buildup but is not the current configuration.

4.3.2 Production Information System: JIT or MRP

The line operates more closely to an MRP-driven environment, for the following reasons:

- Sewing batches are scheduled based on customer orders and cutting-room release plans.
- No component-level kanban triggers are used.
- Material is pre-purchased and staged according to weekly work orders.

Because of this, information flows top-down, in contrast to a Just-In-Time (JIT) system where flow would be based on downstream consumption.

4.3.3 Is a New Approach Reasonable?

Yes. Converting the bottleneck stations (Close Binding + Label and Close Side Seam) into a more balanced configuration is both feasible and impactful. Adding one machine to relieve the first bottleneck and redistributing work across the line would:

- Increase efficiency from 63% to approximately 90%
- Increase output from ~80 units/hour to ~120 units/hour
- Reduce queue length upstream of the bottleneck
- Shorten lead times and stabilize the flow

This makes adopting a more pull-like workflow, or hybrid JIT elements, reasonable once bottleneck relief is implemented.

5. Delivery Analysis

5.1 Performance measures and utilization

The balanced sewing line proposed in Section 4 directly affects delivery performance, because throughput and utilization determine whether the line can meet the forecasted demand for SKUs 1008MWT and 1011HWLS on a consistent basis.

Using the process times in Table A and the capacities summarized in Table B, the original, unbalanced line operated at approximately 80 pieces/hour, while the balanced configuration with two parallel stations at Close Binding + Label and Close Side Seam increased effective capacity to roughly 120 pieces/hour.

To relate these capacities to demand, the average daily demand for each SKU (from the forecasting section) was converted into an hourly required production rate assuming a standard 8-hour production day. Line utilization was then computed using the standard queueing expression $\rho = \mu\lambda$ where λ (lambda) is the required output rate (units/hour) and μ (mu) is the line capacity (units/hour). The results are summarized in Table D: Line Capacity and Utilization by Layout and SKU.

Measure	**1008MWT**	**1011HWLS**	**Notes**
Annual demand	153,263	103,323	From 12-month forecast
Working days per year	250	250	
Daily demand	613	413	
Length of shift	8	8	One production shift
Arrival rate	76.63	51.66	Effective demand rate at the line
Line capacity – current μ_before	80	80	From observed productivity (80 pcs/hr)
Utilization – current ρ_before=λ/μbefore	0.96	0.65	96% and 65%
Line capacity – proposed μ_after	120	120	After adding a second Close Binding + Label station
Utilization – proposed ρ_after=λ/μ_after	0.64	0.43	64% and 43%

Table D

Under the original (unbalanced) configuration, utilization for 1008MWT is very close to 1.0, indicating that the line is operating almost at capacity. In this region, even small disturbances (machine downtime, rework, late material) cause long queues and delivery delays. After balancing the line, utilization for 1008MWT drops to a more stable range (well below 1.0), while utilization for 1011HWLS remains moderate. This confirms that the balanced layout not only increases throughput but also creates buffer capacity that allows the system to absorb short-term variability without missing shipment dates.

A visual comparison of required demand rate vs. available capacity is provided in Figure 5: Required Output vs Line Capacity (Before vs After Balancing), which reinforces that the balanced line comfortably meets the forecasted demand for both SKUs.

SKU	Required hourly rate (λ)	Capacity - current (μ_before)	Capacity - proposed (μ_after)
1008MWT	76.63	80	120
1011HWLS	51.66	80	120

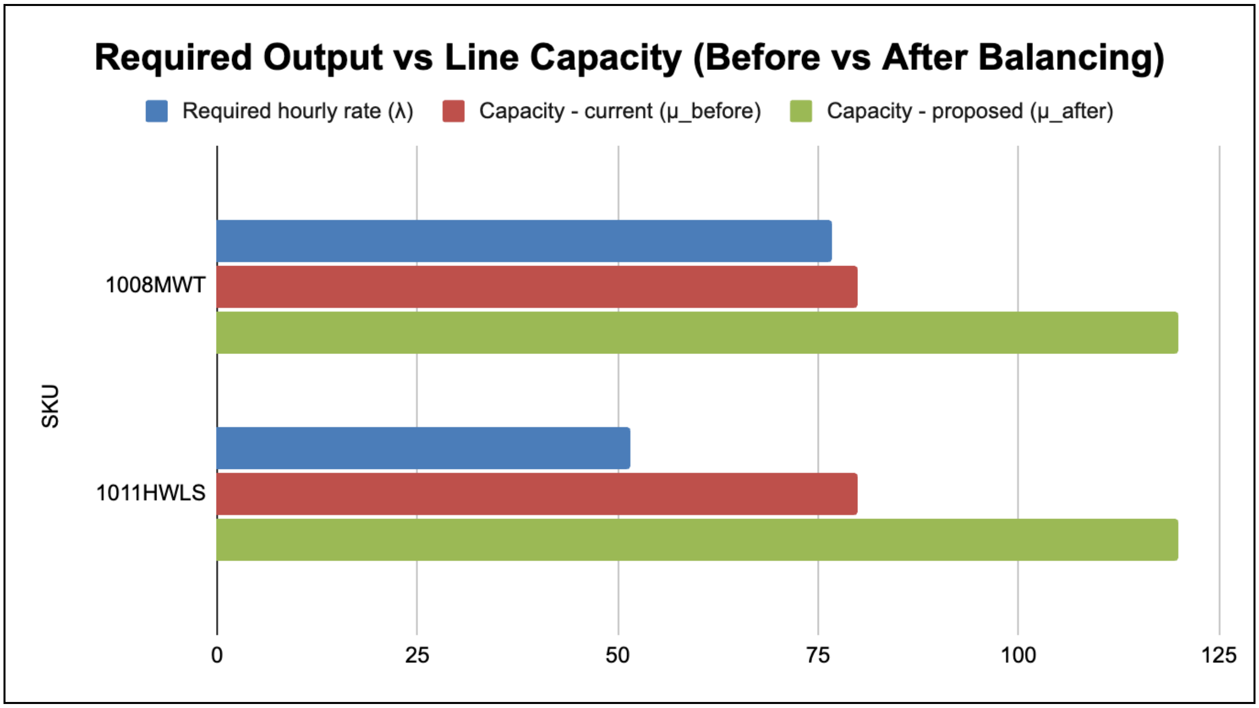


Figure 5

5.2 Logistics issues and bullwhip effect

From a logistics perspective, the products analyzed are basic T-shirts (midweight and heavyweight, short- and long-sleeve). Demand shows seasonal patterns (higher in spring/summer and in Q4 for holiday orders) but is relatively stable within each season. Because the factory receives orders from downstream customers (brands, distributors, or large screen-printing shops), there is still a risk that order variability becomes larger than the underlying consumer demand, this is the classic bullwhip effect.

In this context, the bullwhip effect should be considered but is manageable:

- Forecasts for 1008MWT and 1011HWLS are reasonably smooth at the annual level, but month-to-month orders can spike when customers place large batch orders.
- If House of Blanks only reacts to customer purchase orders and does not see point-of-sale or e-commerce data, production may over-react to short-term changes, causing swings in WIP, overtime, and inventory.
- The new line balance improves the internal flow, but without better information sharing the plant could still experience rush orders followed by idle time.

Therefore, while the main bottleneck is operational (sewing line capacity), logistics and information flow can amplify or dampen that bottleneck depending on how orders are released to the line.

5.3 Role of supply chain coordination

Supply chain coordination can be used to further stabilize deliveries and protect the gains from line balancing. Several coordination mechanisms are relevant for House of Blanks:

- Shared demand information. Providing visibility to downstream demand (e.g., POS data from key customers or aggregated e-commerce trends) would allow the plant to adjust production plans gradually instead of reacting to abrupt order changes.
- Order-smoothing policies. Encouraging customers to place more frequent, smaller orders, aligned with the EOQ and production batch sizes derived in the inventory section, reduces variability and keeps utilization in the desirable range highlighted in Table D.
- Coordinated replenishment cycles. For large repeat customers, agreeing on common replenishment cycles (for example, weekly or bi-weekly) helps synchronize production, inventory, and shipping schedules, reducing the risk of both stockouts and excess finished-goods inventory.

- Priority rules for bottleneck operations. Even with two machines at Close Binding + Label and Close Side Seam, these remain critical resources. Coordinating order release so that these stations always process the highest-priority SKUs (e.g., nearest due date) improves on-time delivery performance without additional capital.

In summary, the balanced sewing line is capable of meeting the forecasted demand for 1008MWT and 1011HWLS, but delivery performance will be strongest when operational improvements are combined with coordinated ordering and information-sharing practices across the supply chain.

Data Appendix

Product ID	Product Name	Date	Units Sold	Production Cost	Selling Price
1008MWT	1008 MIDWEIGHT T-SHIRT	April 2022	11,510	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	April 2023	12,488	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	April 2024	13,577	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	August 2022	11,955	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	August 2023	12,122	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	August 2024	12,310	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	December 2022	13,722	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	December 2023	14,234	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	December 2024	16,249	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	February 2022	11,322	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	February 2023	12,844	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	February 2024	13,910	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	January 2022	10,655	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	January 2023	11,044	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	January 2024	10,833	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	July 2022	12,577	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	July 2023	12,922	$3.90	$30.00

1008MWT	1008 MIDWEIGHT T-SHIRT	July 2024	13,010	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	June 2022	13,199	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	June 2023	13,344	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	June 2024	13,488	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	March 2022	11,990	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	March 2023	12,355	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	March 2024	12,233	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	May 2022	13,244	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	May 2023	13,433	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	May 2024	13,622	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	November 2022	10,588	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	November 2023	10,644	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	November 2024	10,810	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	October 2022	10,955	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	October 2023	10,899	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	October 2024	11,066	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	September 2022	11,399	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	September 2023	11,722	$3.90	$30.00
1008MWT	1008 MIDWEIGHT T-SHIRT	September 2024	12,155	$3.90	$30.00

1009HWT	1009 HEAVYWEIGHT T-SHIRT	April 2022	1,279	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	April 2023	1,186	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	April 2024	1,303	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	August 2022	1,421	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	August 2023	1,430	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	August 2024	1,390	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	December 2022	1,207	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	December 2023	1,169	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	December 2024	1,144	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	February 2022	1,181	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	February 2023	1,217	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	February 2024	1,241	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	January 2022	1,318	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	January 2023	1,214	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	January 2024	1,291	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	July 2022	1,416	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	July 2023	1,445	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	July 2024	1,387	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	June 2022	1,209	$5.30	$35.00

1009HWT	1009 HEAVYWEIGHT T-SHIRT	June 2023	1,174	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	June 2024	1,150	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	March 2022	1,187	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	March 2023	1,229	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	March 2024	1,253	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	May 2022	1,344	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	May 2023	1,234	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	May 2024	1,317	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	November 2022	1,423	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	November 2023	1,450	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	November 2024	1,400	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	October 2022	1,219	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	October 2023	1,193	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	October 2024	1,164	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	September 2022	1,190	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	September 2023	1,234	$5.30	$35.00
1009HWT	1009 HEAVYWEIGHT T-SHIRT	September 2024	1,260	$5.30	$35.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	April 2022	8,420	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	April 2023	8,590	$5.50	$40.00

1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	April 2024	9,574	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	August 2022	7,810	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	August 2023	7,690	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	August 2024	7,880	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	December 2022	8,105	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	December 2023	8,310	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	December 2024	9,050	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	February 2022	9,277	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	February 2023	9,120	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	February 2024	8,450	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	January 2022	8,655	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	January 2023	8,822	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	January 2024	8,033	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	July 2022	7,899	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	July 2023	7,780	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	July 2024	7,988	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	June 2022	8,202	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	June 2023	8,400	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	June 2024	9,763	$5.50	$40.00

1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	March 2022	9,399	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	March 2023	9,222	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	March 2024	8,544	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	May 2022	8,733	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	May 2023	8,899	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	May 2024	8,099	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	November 2022	7,944	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	November 2023	7,810	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	November 2024	8,022	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	October 2022	8,240	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	October 2023	8,488	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	October 2024	9,310	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	September 2022	9,477	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	September 2023	9,255	$5.50	$40.00
1011HWLS	1011 HEAVYWEIGHT L/S T-SHIRT	September 2024	8,610	$5.50	$40.00
1012RFT	1012 RELAXED FIT T-SHIRT	April 2022	604	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	April 2023	615	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	April 2024	642	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	August 2022	666	$4.40	$35.00

1012RFT	1012 RELAXED FIT T-SHIRT	August 2023	674	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	August 2024	654	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	December 2022	563	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	December 2023	555	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	December 2024	547	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	February 2022	553	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	February 2023	561	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	February 2024	574	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	January 2022	610	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	January 2023	622	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	January 2024	653	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	July 2022	676	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	July 2023	684	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	July 2024	662	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	June 2022	575	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	June 2023	564	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	June 2024	555	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	March 2022	560	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	March 2023	570	$4.40	$35.00

1012RFT	1012 RELAXED FIT T-SHIRT	March 2024	579	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	May 2022	614	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	May 2023	626	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	May 2024	660	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	November 2022	680	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	November 2023	689	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	November 2024	665	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	October 2022	585	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	October 2023	576	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	October 2024	566	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	September 2022	570	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	September 2023	582	$4.40	$35.00
1012RFT	1012 RELAXED FIT T-SHIRT	September 2024	591	$4.40	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	April 2022	883	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	April 2023	932	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	April 2024	885	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	August 2022	876	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	August 2023	924	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	August 2024	928	$5.45	$35.00

1016OSBT	1016 OVERSIZED BOXY T-SHIRT	December 2022	901	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	December 2023	841	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	December 2024	855	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	February 2022	969	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	February 2023	872	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	February 2024	982	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	January 2022	852	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	January 2023	949	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	January 2024	959	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	July 2022	891	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	July 2023	948	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	July 2024	935	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	June 2022	847	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	June 2023	983	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	June 2024	863	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	March 2022	990	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	March 2023	906	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	March 2024	919	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	May 2022	931	$5.45	$35.00

1016OSBT	1016 OVERSIZED BOXY T-SHIRT	May 2023	997	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	May 2024	938	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	November 2022	987	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	November 2023	850	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	November 2024	876	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	October 2022	917	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	October 2023	981	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	October 2024	886	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	September 2022	964	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	September 2023	914	$5.45	$35.00
1016OSBT	1016 OVERSIZED BOXY T-SHIRT	September 2024	898	$5.45	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	April 2022	1,447	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	April 2023	1,483	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	April 2024	1,536	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	August 2022	1,616	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	August 2023	1,650	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	August 2024	1,586	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	December 2022	1,347	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	December 2023	1,327	$2.30	$35.00

1017SPT	1017 SLUB POCKET T-SHIRT	December 2024	1,307	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	February 2022	1,323	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	February 2023	1,356	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	February 2024	1,400	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	January 2022	1,463	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	January 2023	1,510	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	January 2024	1,563	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	July 2022	1,643	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	July 2023	1,680	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	July 2024	1,600	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	June 2022	1,376	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	June 2023	1,340	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	June 2024	1,320	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	March 2022	1,343	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	March 2023	1,380	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	March 2024	1,413	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	May 2022	1,480	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	May 2023	1,524	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	May 2024	1,573	$2.30	$35.00

1017SPT	1017 SLUB POCKET T-SHIRT	November 2022	1,657	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	November 2023	1,690	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	November 2024	1,613	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	October 2022	1,390	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	October 2023	1,353	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	October 2024	1,326	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	September 2022	1,346	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	September 2023	1,380	$2.30	$35.00
1017SPT	1017 SLUB POCKET T-SHIRT	September 2024	1,423	$2.30	$35.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	April 2022	519	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	April 2023	536	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	April 2024	558	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	August 2022	434	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	August 2023	416	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	August 2024	403	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	December 2022	625	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	December 2023	616	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	December 2024	597	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	February 2022	588	$18.70	$110.00

3008PHS	3008 PULLOVER HOODED SWEATSHIRT	February 2023	575	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	February 2024	564	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	January 2022	528	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	January 2023	550	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	January 2024	571	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	July 2022	440	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	July 2023	422	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	July 2024	409	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	June 2022	431	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	June 2023	418	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	June 2024	405	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	March 2022	517	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	March 2023	502	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	March 2024	489	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	May 2022	508	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	May 2023	521	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	May 2024	534	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	November 2022	603	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	November 2023	618	$18.70	$110.00

3008PHS	3008 PULLOVER HOODED SWEATSHIRT	November 2024	603	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	October 2022	500	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	October 2023	487	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	October 2024	476	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	September 2022	491	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	September 2023	501	$18.70	$110.00
3008PHS	3008 PULLOVER HOODED SWEATSHIRT	September 2024	515	$18.70	$110.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	April 2022	260	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	April 2023	269	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	April 2024	284	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	August 2022	218	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	August 2023	209	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	August 2024	202	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	December 2022	315	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	December 2023	307	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	December 2024	298	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	February 2022	291	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	February 2023	282	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	February 2024	275	$21.35	$115.00

5002ZHS	5002 ZIP HOODED SWEATSHIRT	January 2022	262	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	January 2023	278	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	January 2024	293	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	July 2022	222	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	July 2023	211	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	July 2024	204	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	June 2022	220	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	June 2023	209	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	June 2024	203	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	March 2022	258	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	March 2023	249	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	March 2024	240	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	May 2022	253	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	May 2023	260	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	May 2024	271	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	November 2022	302	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	November 2023	311	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	November 2024	300	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	October 2022	247	$21.35	$115.00

5002ZHS	5002 ZIP HOODED SWEATSHIRT	October 2023	238	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	October 2024	231	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	September 2022	242	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	September 2023	252	$21.35	$115.00
5002ZHS	5002 ZIP HOODED SWEATSHIRT	September 2024	262	$21.35	$115.00
7004CSP	7004 CLASSIC SWEATPANT	April 2022	333	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	April 2023	343	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	April 2024	364	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	August 2022	275	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	August 2023	266	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	August 2024	260	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	December 2022	404	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	December 2023	398	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	December 2024	391	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	February 2022	383	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	February 2023	376	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	February 2024	368	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	January 2022	338	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	January 2023	349	$15.25	$110.00

7004CSP	7004 CLASSIC SWEATPANT	January 2024	362	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	July 2022	280	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	July 2023	269	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	July 2024	262	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	June 2022	273	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	June 2023	265	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	June 2024	260	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	March 2022	328	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	March 2023	319	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	March 2024	313	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	May 2022	322	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	May 2023	329	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	May 2024	338	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	November 2022	400	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	November 2023	407	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	November 2024	399	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	October 2022	318	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	October 2023	311	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	October 2024	304	$15.25	$110.00

7004CSP	7004 CLASSIC SWEATPANT	September 2022	312	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	September 2023	320	$15.25	$110.00
7004CSP	7004 CLASSIC SWEATPANT	September 2024	328	$15.25	$110.00
7007FSS	7007 FLEECE SWEATSHORT	April 2022	413	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	April 2023	433	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	April 2024	486	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	August 2022	633	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	August 2023	659	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	August 2024	613	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	December 2022	273	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	December 2023	260	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	December 2024	246	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	February 2022	253	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	February 2023	265	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	February 2024	280	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	January 2022	419	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	January 2023	453	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	January 2024	506	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	July 2022	653	$9.50	$65.00

7007FSS	7007 FLEECE SWEATSHORT	July 2023	673	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	July 2024	626	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	June 2022	288	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	June 2023	273	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	June 2024	258	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	March 2022	267	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	March 2023	280	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	March 2024	293	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	May 2022	427	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	May 2023	446	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	May 2024	493	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	November 2022	659	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	November 2023	680	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	November 2024	640	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	October 2022	312	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	October 2023	299	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	October 2024	283	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	September 2022	293	$9.50	$65.00
7007FSS	7007 FLEECE SWEATSHORT	September 2023	306	$9.50	$65.00

7007FSS	7007 FLEECE SWEATSHORT	September 2024	320	$9.50	$65.00

Business Data by Product

Product Name	Total Units Sold (2022-2024)	Total Production Cost (2022-2024)	Total Revenue (2022-2024)	Total Profit (2022-2024)	Profit Total
1008 MIDWEIGHT T-SHIRT	444,430	$1,733,277.000	$13,332,900.000	$11,599,623.000	36.86996886
1011 HEAVYWEIGHT L/S T-SHIRT	307,870	$1,693,285.000	$12,314,800.000	$10,621,515.000	33.76100476
3008 PULLOVER HOODED SWEATSHIRT	18,470	$345,389.000	$2,031,700.000	$1,686,311.000	70.63097361
1012 RELAXED FIT T-SHIRT	21,922	$96,456.800	$767,270.000	$670,813.200	
7004 CLASSIC SWEATPANT	11,867	$180,971.750	$1,305,370.000	$1,124,398.250	
1016 OVERSIZED BOXY T-SHIRT	33,029	$180,008.050	$1,156,015.000	$976,006.950	
1009 HEAVYWEIGHT T-SHIRT	45,869	$243,105.700	$1,605,415.000	$1,362,309.300	
1017 SLUB POCKET T-SHIRT	52,764	$121,357.200	$1,846,740.000	$1,725,382.800	
5002 ZIP HOODED SWEATSHIRT	9,228	$197,017.800	$1,061,220.000	$864,202.200	
7007 FLEECE SWEATSHORT	14,961	$142,129.500	$972,465.000	$830,335.500	
				$31,460,897.200	

Profit Calculations

	Month	Actual Demand	Level (Lt)	Trend (Tt)	Seasonal (St)	Forecast (Ft)	Error	Error²	Error %		Lo	
2022	January	10,655			0.878						11,926	
	February	11,322			1.028							
	March	11,990			0.988						To	
	April	11,510			1.015						-36.194	
	May	13,244			1.088							
	June	13,199			1.081						α	
	July	12,577			1.040						0.3	
	August	11,955			0.982						β	
	September	11,399			0.952						0.1	
	October	10,955			0.889						γ	
	November	10,588			0.865						0.3	
	December	13,722			1.194							
2023	January	11,044	11,926.333	-36.194	0.893					MAD	MSE	MAPE
	February	12,844	12,071.035	-18.105	1.039	12,224.078	620	384302.855	0.048	126	232731.613	0.024
	March	12,355	12,189.944	-4.403	0.995	11,903.933	451	203461.563	0.037	103	225842.011	0.023
	April	12,488	12,222.521	-0.705	1.017	12,362.937	125	15640.703	0.010	87	226907.747	0.022
	May	13,433	12,258.854	2.998	1.090	13,298.660	134	18047.326	0.010	85	237471.099	0.022
	June	13,344	12,286.973	5.510	1.082	13,253.495	91	8191.198	0.007	82	249019.719	0.023
	July	12,922	12,333.038	9.566	1.042	12,781.439	141	19757.474	0.011	82	262399.081	0.024
	August	12,122	12,341.269	9.432	0.982	12,126.373	-4	19.127	0.000	78	276672.117	0.025
	September	11,722	12,337.526	8.115	0.952	11,763.833	-42	1749.964	0.004	84	293962.929	0.026
	October	10,899	12,320.444	5.595	0.888	10,973.656	-75	5573.468	0.007	92	313443.793	0.028
	November	10,644	12,319.096	4.901	0.865	10,664.022	-20	400.871	0.002	104	335434.530	0.029
	December	14,234	12,204.457	-7.053	1.185	14,709.600	-476	226195.153	0.033	113	361206.350	0.032
2024	January	10,833	12,178.792	-8.914	0.892	10,888.381	-55	3067.036	0.005	162	372457.284	0.031
	February	13,910	12,535.778	27.676	1.060	12,642.923	1,267	1605483.021	0.091	182	406038.215	0.034
	March	12,233	12,481.245	19.455	0.991	12,505.772	-273	74404.729	0.022	74	286093.735	0.028
	April	13,577	12,756.660	45.051	1.031	12,709.545	867	752477.382	0.064	112	309614.735	0.029
	May	13,622	12,708.963	35.776	1.085	13,959.109	-337	113642.602	0.025	18	254256.904	0.024
	June	13,488	12,659.611	27.263	1.077	13,795.148	-307	94339.693	0.023	68	274344.662	0.024
	July	13,010	12,625.882	21.164	1.039	13,221.882	-212	44894.141	0.016	131	304345.490	0.024
	August	12,310	12,612.068	17.666	0.980	12,424.540	-115	13119.459	0.009	200	356235.760	0.026
	September	12,155	12,672.093	21.902	0.954	12,020.614	134	18059.558	0.011	278	442014.835	0.030
	October	11,066	12,626.010	15.104	0.884	11,267.145	-201	40459.184	0.018	326	583333.261	0.037
	November	10,810	12,598.690	10.861	0.863	10,932.298	-122	14956.921	0.011	590	854770.300	0.046
	December	16,249	12,939.002	43.806	1.207	14,947.238	1,302	1694583.678	0.080	1,302	1694583.678	0.080

1008MWT Forecast

	Month	Actual Demand	Level (Lt)	Trend (Tt)	Seasonal (St)	Forecast (Ft)	Error	Error²	Error %		Lo	
2022	January	8,655			0.99						8,513	
	February	9,277			1.05							
	March	9,399			1.06						To	
	April	8,420			1.04						-6.507	
	May	8,733			1.00							
	June	8,202			1.03						α	
	July	7,899			0.92						0.3	
	August	7,810			0.91						β	
	September	9,477			1.07						0.1	
	October	8,240			1.01						γ	
	November	7,944			0.93						0.3	
	December	8,105			0.99							
2023	January	8,822	11,926.333	-36.194	0.918					MAD	MSE	MAPE
	February	9,120	10,937.704	-131.438	0.983	12,442.182	-3,322	11036896.104	0.364	-1	1255124.628	0.090
	March	9,222	10,177.287	-194.336	1.013	11,441.929	-2,220	4928086.053	0.241	150	810498.652	0.078
	April	8,590	9,475.091	-245.122	0.997	10,344.110	-1,754	3076901.793	0.204	263	614423.061	0.070
	May	8,899	9,122.879	-255.831	0.995	9,257.011	-358	128171.898	0.040	364	491299.125	0.064
	June	8,400	8,659.151	-276.621	1.010	9,112.147	-712	507154.009	0.085	402	510411.084	0.065
	July	7,780	8,397.907	-275.083	0.924	7,732.719	47	2235.504	0.006	464	510592.033	0.064
	August	7,690	8,217.542	-265.611	0.919	7,402.278	288	82783.762	0.037	488	540495.358	0.067
	September	9,255	8,171.627	-243.642	1.086	8,474.552	780	609098.382	0.084	501	569102.333	0.069
	October	8,488	8,058.615	-230.579	1.026	8,046.080	442	195293.723	0.052	482	566435.930	0.068
	November	7,810	8,007.873	-212.595	0.941	7,254.467	556	308616.611	0.071	485	592946.087	0.069
	December	8,310	7,968.377	-195.285	1.008	7,737.296	573	327990.174	0.069	479	614817.585	0.069
2024	January	8,033	8,066.519	-165.942	0.941	7,135.181	898	806079.426	0.112	472	638719.870	0.069
	February	8,450	8,110.178	-144.982	1.000	7,763.454	687	471345.675	0.081	433	623505.364	0.065
	March	8,544	8,105.900	-130.912	1.025	8,068.880	475	225739.320	0.056	407	638721.333	0.063
	April	9,574	8,462.466	-82.164	1.038	7,953.461	1,621	2626146.301	0.169	400	684608.224	0.064
	May	8,099	8,308.885	-89.306	0.989	8,335.792	-237	56070.594	0.029	247	441915.964	0.051
	June	9,763	8,652.543	-46.009	1.046	8,304.822	1,458	2126282.347	0.149	317	497036.731	0.054
	July	7,988	8,619.027	-44.760	0.925	7,949.533	38	1479.710	0.005	126	225495.795	0.038
	August	7,880	8,575.338	-44.653	0.919	7,876.721	3	10.755	0.000	144	270299.012	0.045
	September	8,610	8,350.417	-62.680	1.069	9,262.439	-652	425676.978	0.076	179	337871.077	0.056
	October	9,310	8,522.546	-39.199	1.046	8,506.632	803	645399.543	0.086	456	308602.443	0.050
	November	8,022	8,495.027	-38.031	0.942	7,985.351	37	1343.167	0.005	282	140203.893	0.031
	December	9,050	8,614.272	-22.303	1.021	8,521.734	528	279064.618	0.058	528	279064.618	0.058

1011HWLS Forecast

References

Nahmias & Olsen (Production and Operations Analytics)
Nahmias, S., & Olsen, T. L. (2020). *Production and operations analytics* (8th ed.). Waveland Press.

Goldratt & Cox (The Goal)
Goldratt, E. M., & Cox, J. (2004). *The goal: A process of ongoing improvement* (3rd rev. ed.). North River Press.

Notes

“Progress in understanding requires that we challenge basic assumptions about how the world is and why it is that way. If we can better understand our world and the principles that govern it, I suspect all our lives will be better.”

-Eliyahu M. Goldratt